Geography of Colombian Life Satisfaction

Thursday 16th July, 2026 12:05

Happiness, Colombia, Encuesta Nacional de Calidad de Vida

One of the puzzles in the happiness literature is Latin American phenomenon (Rojas 2015, 2019): high happiness[1] despite low development in Latin America. Colombia is one of the happiest countries in the world–see table 1. But we know little about geography of happiness within Colombia. This paper is about geography of Colombian happiness across its geographies: departments/provinces, cities/municipalities, and urban-rural and non-farmers/farmers. But first we start with overall physical and social geography of Colombia to provide background.

Table 1: 10 happiest countries in the world. Data from World Database of Happiness (WDH) 2010-2019 out of 160 countries at `worlddatabaseofhappiness.eur.nl/rank-reports/satisfaction-with-life`; and World Values Surveys (WVS) 2005-2022 [waves 5-7] out of 88 countries at `https://www.worldvaluessurvey.org`. Technical information for WDH is at `https://worlddatabaseofhappiness.eur.nl/reports/finding-reports-on-happiness-in-nations/technical-details-to-rank-reports-of-happiness-in-nations/technical-details-to-rank-report-average-happiness-in-nations-2010-2019` and WVS documentation is at `https://www.worldvaluessurvey.org/WVSDocumentationWV7.jsp`.

WDH			WVS		
rank	country	happiness (1-10)	rank	country	happiness (1-10)
1	Denmark	8.2	1	Puerto Rico	8.4
2	Mexico	8.1	2	Mexico	8.3
3	Colombia	8.1	3	Colombia	8.3
4	Switzerland	8	4	Qatar	8.0
5	Finland	8	5	Norway	7.9
6	Iceland	8	6	Nicaragua	7.9
7	Costa Rica	7.9	7	Tajikistan	7.9
8	Norway	7.9	8	Switzerland	7.9
9	Canada	7.9	9	Uzbekistan	7.9
10	Qatar	7.8	10	Ecuador	7.8

1 Colombia's Geography

Here we mostly focus on human/social geography. Physical geography is mostly postponed to appendix.

Colombia is at North-West of South America (fig 1), a large country of about 54m, third largest in Latin America after Brazil and Mexico (`https://fred.stlouisfed.org/series/POPTOTCOA647NWDB`.) Colombia is also large in land mass at about 1.1m sq km, while much smaller than Brazil at 8.5m sq km, it is larger than Venezuela at .9m sq km, and about 3x larger than Germany at .35m sq km and about 2x of Spain at .5 sq km.

¿¿¿then about dept guess mv that map with names from app, and cities (like cp that table from unhappy bog maybe or to app or some map of largest cities better yet)

[1] Following usual practice in social indicators research terms happiness and life satisfaction are used interhcangeably, and specifically we mean cognitive and evaluative life satisfaction rather than affective happiness.

Figure 1: Colombia circled in red in Latin America shaded in green.

¿¿¿first just write myself; then add from Urbanism in Colombia in /papers/root/old/2025/swbColCitReg/tex/swbColCitReg.tex about colombia boilerplate like 2nd biggest land mas and pop in lat am about roads etc ranked one of thw worst and airports

Colombia has extraordinary diversity, not just biodiversity (again 2nd largest in the world after Brazil), but also cultural and social. For instance, Bogota (or at least parts of it) is simiar to other large fast paced commerce oriented cities like New York or London. Likewise, it is very international and has the most connected airport in Latin America (most international connections) and one of the 3 largest airports in Latin America (close to Sao Paulo and Mexico city).[2] Cartagena is Miami-like: expensive and fashionable catering to the rich and tourists. On the other hand, Cali, the third largest city, is very much Latin with more laid back atmosphere and salsa music. Indeed, one could think these cities belong to different countries or even cultures. Pacific Buenaventura and Tumaco are different yet–less organized and impoverished. One could enumerate many more examples of differences across municipalities.

add from ebshoy paper on primate city Bogota could be considered a primate city, although the difference versus second largest Medellin is only of order of about 2, smaller than in many other countries[3]

2 Geography of Colombian happiness–the Literature

There are only few studies on the topic.

We know little about colombian happiness: my paper, stiupid burger as cited in my paper, and that govt report i cite in my paper!! do skim thru it!!

burger few observations: only few thousand (7-9k depending on model) and problematic measurement across space as elaborated in urb unhappiness is common

and there is interesting (Peiró-Palomino et al. 2020) but it uses an index objective indicators of quality of life such as income, housing and pollution.

There is a recent United Nations study on Colombian happiness using the same Encuesta Nacional de Calidad de Vida as we use, just earlier waves (PNUD 2023). The study is availablee from the the UN[4]

It is closest to our study and most useful overview of Colombian happiness geography to date. There is in fact a similar map to ours in Figure 1.9 on page 26 for 2018-2021, but it is difficult to read as the colormap is undistinctive. While happy Cafetero agrees with our analysis, happy Narino does not. Likewise it finds Guaviare to be unhappy, while we find it moderately happy. A possible explanation for the differences is different timeframe: 2018-2021 theirs v 2023 and 2021 ours. Likewise in figure 1.10 they break down

[2]https://www.oag.com/blog/latin-americas-most-connected-airports-aviation-infographics and https://en.wikipedia.org/wiki/List_of_the_busiest_airports_in_Latin_America.

[3]https://en.wikipedia.org/wiki/Primate_city

[4]https://www.undp.org/sites/g/files/zskgke326/files/2023-02/undp_co_pub_Percepciones_bienestar_subjetivo_2do_cuaderno_INDH_Feb21_2023.pdf.

SWB by urban-rural, ours agrees that there is almost no difference, and in addition to urban-rural breakdown we add farmer-nonfarmer breakdown and add income breakdown for both.

3 Data

We use a great source of data ECV. We use 2023 edition. (and in appendix for robustness check 2021). Sampling details are at https://microdatos.dane.gov.co/index.php/catalog/734

The ECV is nationally representative, cross-sectional annual survey in Colombia, it includes about 250k persons and 80k households across rural and urban.[5]

It is representative of departments, but not municipalities, hence municipalities results are provisional.

The life satisfaction item reads: ?? and rural and farmer and money percapita

4 Departments/Provinces

there are 166k missing out of 242k, we conduct robustness check in appendix using 2021 wave

the map is below, table including number of observations per province is in appendix

unhappy: West (poor Choco, and Southern Cauca and Narino) and East Guainia and Vichada. In 2021 (see appendix) Choco and Cauca were still relatively unhappy, but slightly happier.

Happy Cafetero: Caldas, Risaralda, Quindio–in 2021 there were slight differences (see appendix) but overall Cafetero was happy too. And also happy Santander. Then slightly less happy Norte de Santander, Cesar and Atlantico. In 2021 (see appendix) Santander, Cesar and Atlantico were less happy.

[5] https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-ecv-2021

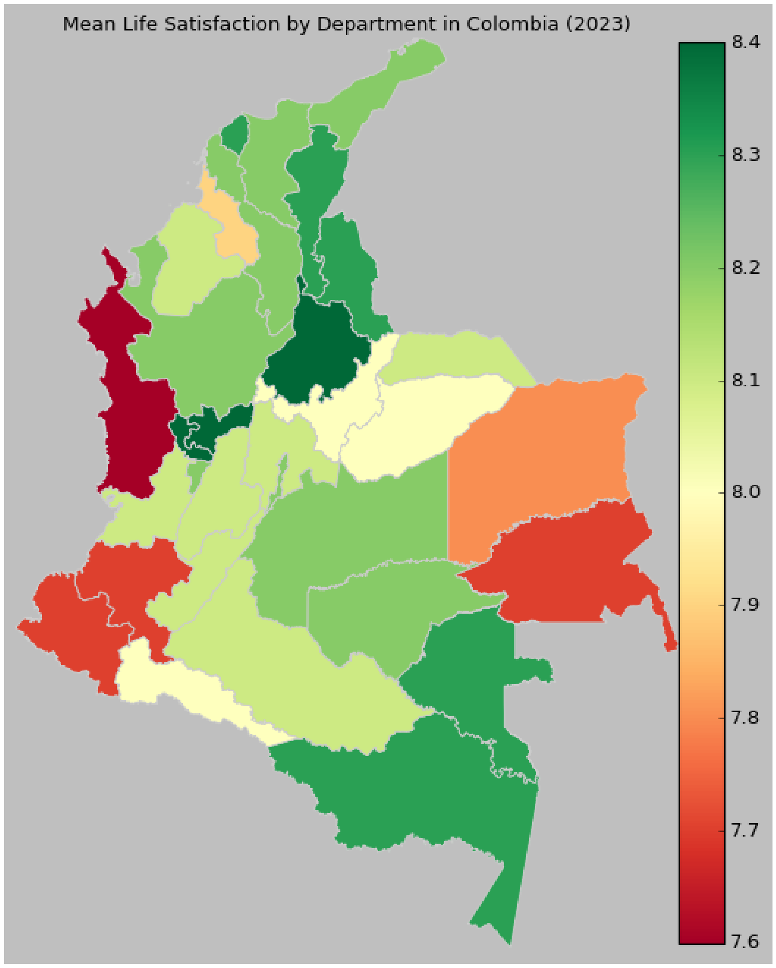


Figure 2: Similar to 2021 robustness/sensitivity check in appendix except: small sample Amazonas and Vaupés, both South-Eastern at 8.3 in 2023 v 7.6 in 2021, a .7 difference. Sencond largest differences: Santander and Putumayo increased by .4 from 2021 to 2023.

relatively unhappy caraibean cordoba and sucre, but then bolivar where caratgena is located is quite happy and so Atlantico where is Barranquilla, at least in 2023 (not so much in 2021).

5 Cities/Municipalities

166k/242k missing, again robustness/sensitivity check in appendix with 2021 ecv

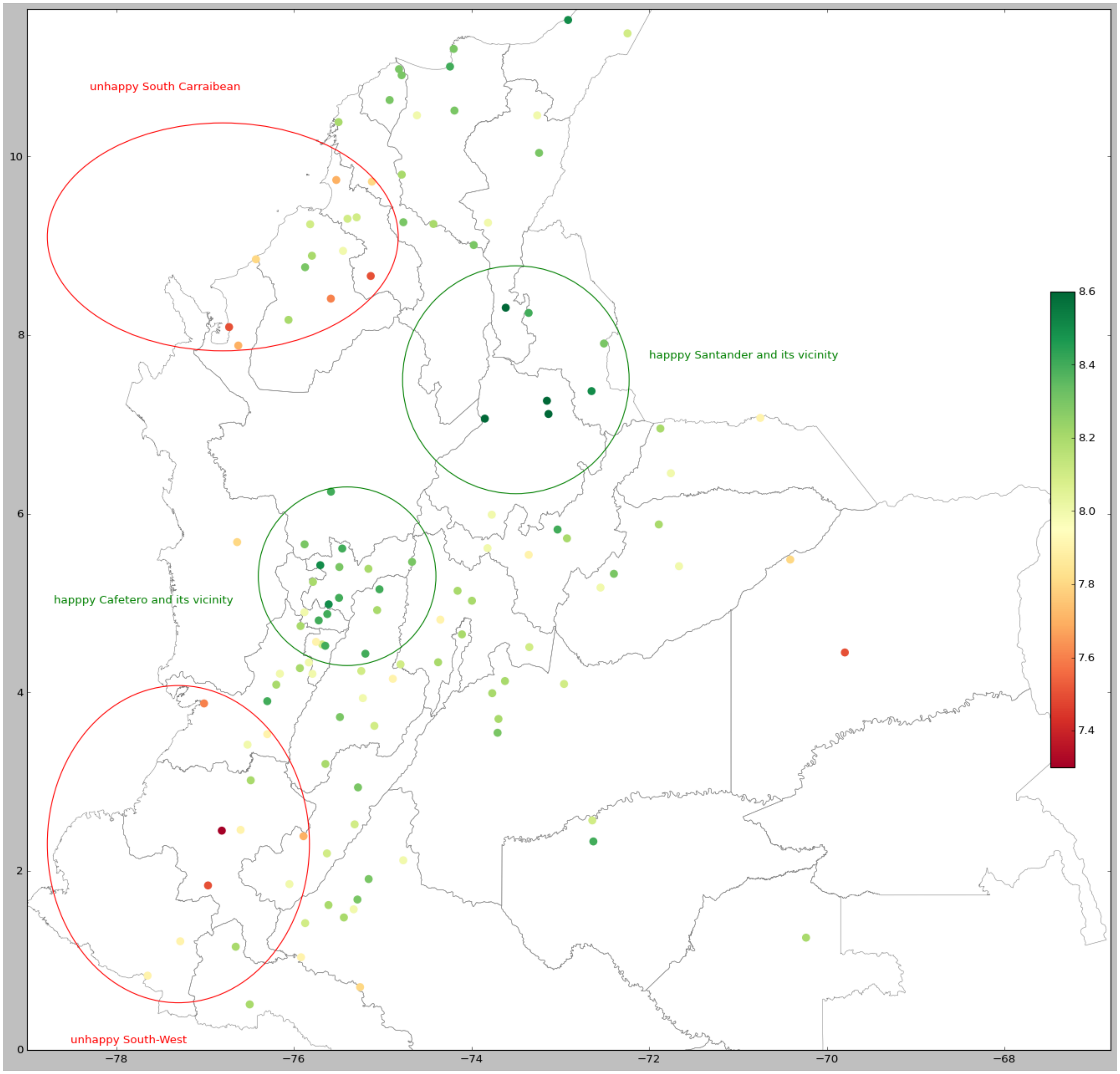


Figure 3: Zoom in to see the detail

Santander is one of the happiests departments, adn so 2 santander largest cities shine: Bucaramanga and Barrancabermeja at 8.6. Bucaramanga was less happy in 2021 at 8.1. Another happy region Cafetero has many happy cities: Calarcá, Pereira, Manizales, and Armenia (Armenia was even happier in 2021). Other happy cities include 2nd largest Medellin, Ibague, and Ciénaga (Ciénaga was less happy in 2021 at 7.9). The least happy is remote Western Puerto Carreño at 7.5, and several cities at 7.9: Tunja, and Southern cities: Palmira in Valle de Cauca, Popayan in Cauca, and Pasto in Narino.

in 2021 results are similar, except that santander and its vicinity is less happy

6 Urban-Rural and Farmers

While most developed countries cities are unhappy and indeed unhappy city is a common pattern across the world (Okulicz-Kozaryn and Valente 2021), it is not always the case in devloping countries.

here boilerplate from happy col unahhpy bog: that cities are growth engines Glaeser (2011) and O'Sullivan (2009) and developing countries like colombia do need economic growth hence role for cities is gretaer than in already develpd countries

and there is paradox of happy peasant and unhappy millionaire Graham (2009)

though not so much in colombia, urban and rural are about same happy, except by income: bar chart by income category bu urb rur as in raman

Both rural and farmer in figures **??** **??**–there are slight differences, but overall similar pattern–no difference for the bottom and 9th deciles; rural and farmer happier in 2nd-8th deciles, and urban happier in the top decile. This makes sense, one needs money to enjoy city, where costof living is higher. In addition, life is more commodified in the city–one needs to pay for more things. For instance, many in the rural areas grow their own food. The bottom decile parity could be explained as that there is some minimum income necessary, and below it, it doesnt help to live in rural v urban area.

In other words, there are lower returns in happiness from income for rural (farmers) v urban persons (non-rarmers). Or Bigger premium from income in urban

Overall, it is worth emphasizing that except the 1st decile and up to median and one decile above it–farmers/rural clearly happier.

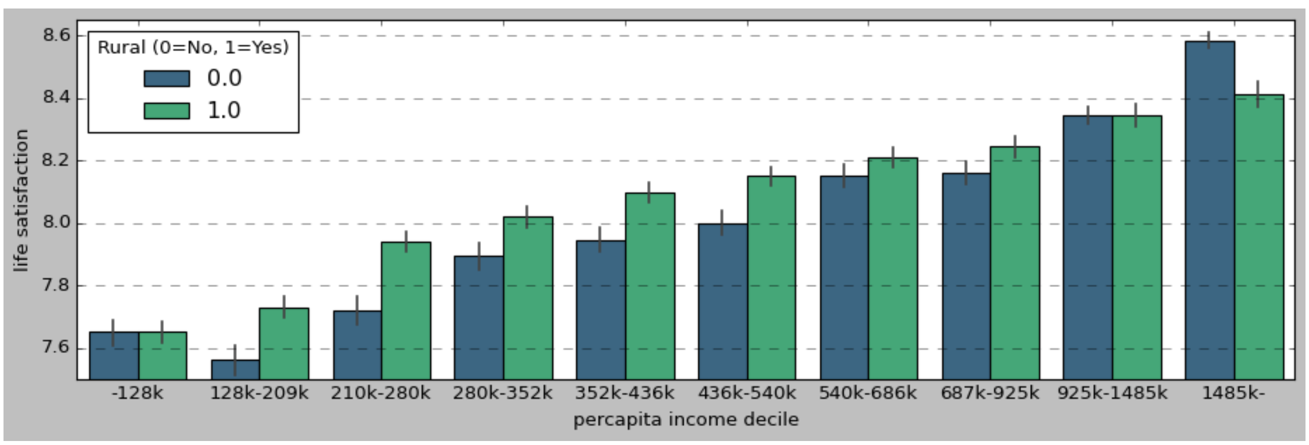


Figure 4

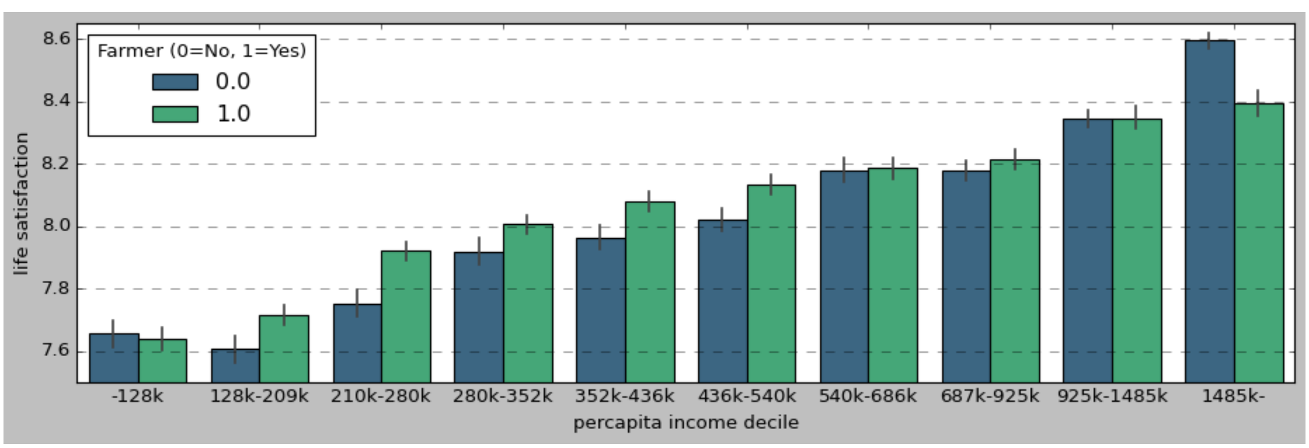


Figure 5

For rural in 2021 (appendix) pattern is similar but less strong for lower deciles where rural are happier than urban, but stronger for 2 top deciles where the opposite is true. For farmers in 2021 there is no happiness premium in lower deciles, but for non-farmers there is happiness premium in 2 top deciles.

ONLINE APPENDIX (Supplementary Online Material (SOM)

[note: this section will NOT be a part of the final version of the manuscript, but will be available online instead]

7 Reeestimation Using 2023 ECV

2021 May be biased say bc of covid, especially that it peaked in colombia in 2021 `https://coronavirus.jhu.edu/region/colombia`(by 2023 used in the paper it pretty much cleared). But it a useful robustness check, if covid didnt change much , then it should be pretty stable

and it was quite high mortality in col `https://coronavirus.jhu.edu/data/mortality`

7.1 Departments/Provinces

Again as said in the paper, overall results are similar to 2023 reported in the body of the paper, except both small sample Amazonas and Vaupés, South-Eastern and with 8.3 in 2023 v 7.6 in 2021, a .7 difference. Sencond largest differences are for Santander and Putumayo which increased by .4 from 2021 to 2023.

Table 2: Life satisfaction by department.

department	life satisfaction 2023	total obs 2023	non-missing life satisfaction 2023	mean life satisfaction 2021	total obs 2021	non missing life satisfaction 2021	2023-2021
Amazonas	8.3	276	245	7.6	423	365	0.7
Antioquia	8.2	5499	4729	8.3	6036	5060	-0.1
Arauca	8.1	974	773	8.1	1103	868	0
Atlántico	8.3	3144	2311	8.0	2338	1799	0.3
Bogotá, D.C.	8.2	4688	3446	8.1	5337	3810	0.1
Bolívar	8.2	2584	2258	8.1	2495	2156	0.1
Boyacá	8.0	4129	3649	7.8	4638	4122	0.2
Caldas	8.4	2822	2555	8.2	3335	3005	0.2
Caquetá	8.1	2547	2018	7.9	2846	2249	0.2
Casanare	8.0	1749	1370	8.1	2051	1457	-0.1
Cauca	7.7	2580	2244	7.9	2572	2287	-0.2
Cesar	8.3	2224	1717	8.0	1833	1510	0.3
Chocó	7.6	857	754	7.9	1016	913	-0.3
Cundinamarca	8.1	3818	3322	7.8	4233	3684	0.3
Córdoba	8.1	2706	2240	8.0	2226	1933	0.1
Guainía	7.7	327	273	7.4	350	274	0.3
Guaviare	8.2	700	457	8.0	621	411	0.2
Huila	8.1	2961	2322	8.0	2999	2407	0.1
La Guajira	8.2	951	750	7.9	988	736	0.3
Magdalena	8.2	2458	2096	8.0	2110	1710	0.2
Meta	8.2	3683	2752	8.1	3862	2817	0.1
Nariño	7.7	2284	2034	7.4	2847	2492	0.3
Norte de Santander	8.3	2499	2028	8.2	2587	2171	0.1
Putumayo	8.0	950	752	7.6	1272	1012	0.4
Quindío	8.2	2085	1658	8.4	2186	1785	-0.2
Risaralda	8.4	1634	1353	8.3	1996	1663	0.1
San Andrés y Providencia	8.4	28	25	8.2	29	26	0.2
Santander	8.4	3790	3194	8.0	4153	3574	0.4
Sucre	7.9	1798	1399	7.8	1899	1467	0.1
Tolima	8.1	3564	3215	8.0	4257	3770	0.1
Valle del Cauca	8.1	4892	4093	8.1	5268	4421	0
Vaupés	8.3	265	245	7.6	288	267	0.7
Vichada	7.8	515	455	7.6	584	482	0.2

7.2 Cities/Municipalities

Very important this robustness check because ECV is not representative of municipalities.

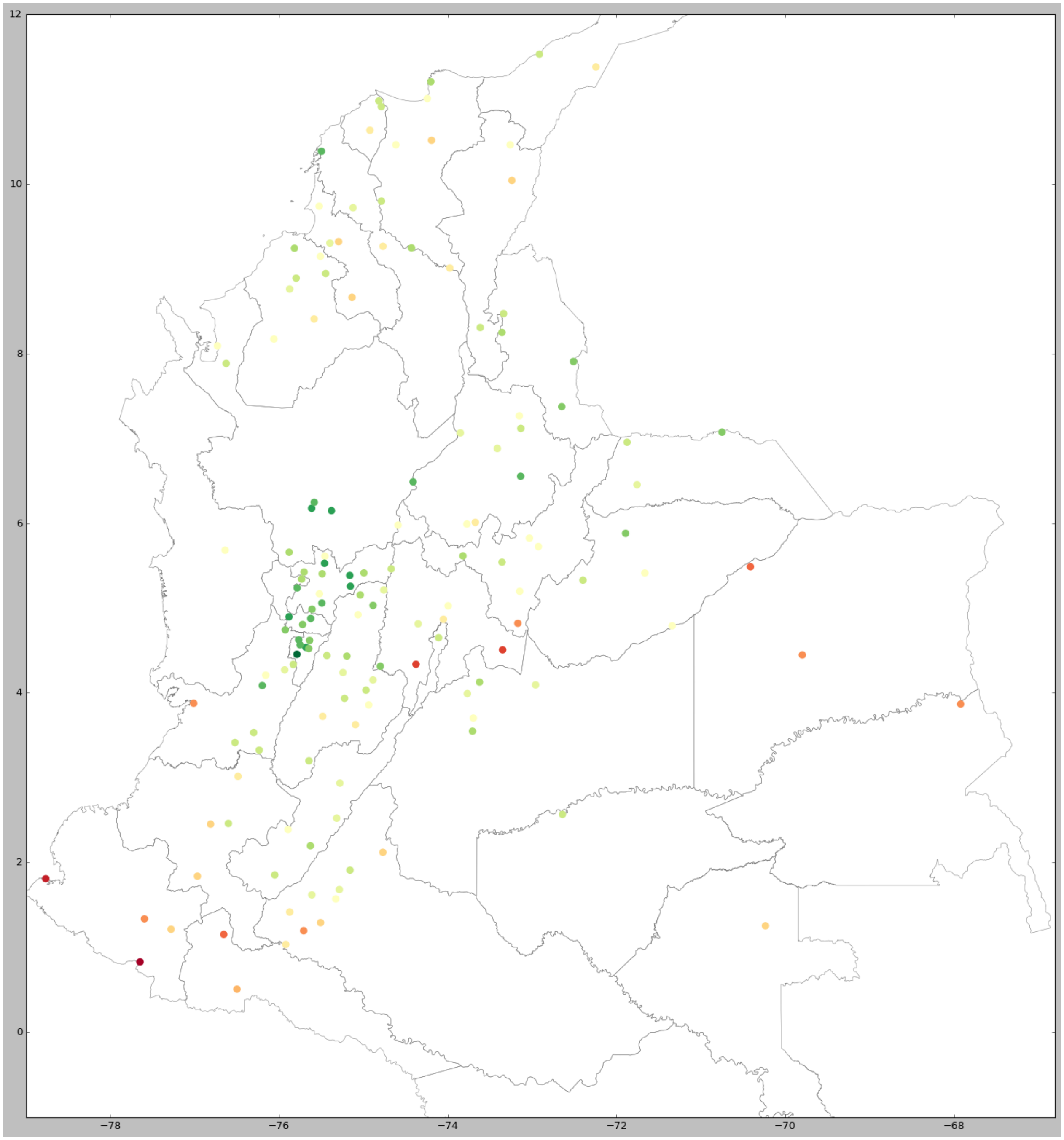


Figure 6: Zoom in to see the detail

Table 3: Life satisfaction by municipality, where non-missing on life satisfaction ¿200 or 220??

municipality name	mean life satisfaction 2023	non missing life satisfaction 2023	mean life satisfaction 2021	non missing life satisfaction 2021	2023-2021 diff
Buga	NaN	NaN	8.0	215	NaN
Espinal	NaN	NaN	7.8	220	NaN
Mariquita	NaN	NaN	7.9	201	NaN
Ocaña	NaN	NaN	8.2	200	NaN
San Vicente del Caguán	NaN	NaN	8.1	240	NaN
Armenia	8.1	668.0	8.5	681	-0.4
Buenaventura	8.1	412.0	8.3	279	-0.2
Cartagena	8.2	591.0	8.4	572	-0.2
Facatativá	8.1	200.0	8.3	222	-0.2
Palmira	7.9	319.0	8.1	332	-0.2
Popayán	7.9	225.0	8.1	232	-0.2
Tuluá	8.2	303.0	8.4	307	-0.2
Tunja	7.9	270.0	8.1	241	-0.2
Cali	8.0	985.0	8.1	1108	-0.1
Pitalito	8.0	293.0	8.1	288	-0.1
Cúcuta	8.3	573.0	8.3	553	0.0
Manizales	8.4	462.0	8.4	567	0.0
Medellín	8.4	905.0	8.4	959	0.0
Puerto Carreño	7.5	225.0	7.5	215	0.0
San José del Guaviare	8.1	246.0	8.1	231	0.0
Villavicencio	8.2	982.0	8.2	990	0.0
Acacías	8.3	229.0	8.2	233	0.1
Bogotá, D.C.	8.2	3446.0	8.1	3836	0.1
Calarcá	8.4	244.0	8.3	257	0.1
Pereira	8.4	524.0	8.3	677	0.1
Santa Marta	8.3	225.0	8.2	218	0.1
Sincelejo	8.1	222.0	8.0	275	0.1
Valledupar	8.0	339.0	7.9	274	0.1
Barranquilla	8.3	1363.0	8.1	1116	0.2
Florencia	8.2	575.0	8.0	630	0.2
Ibagué	8.4	478.0	8.2	549	0.2
Pasto	7.9	244.0	7.7	309	0.2
Yopal	8.3	264.0	8.1	240	0.2
Montería	8.3	465.0	8.0	372	0.3
Neiva	8.3	417.0	8.0	436	0.3
Riosucio	8.5	203.0	8.2	203	0.3
Sogamoso	8.2	236.0	7.9	367	0.3
Aguachica	8.6	202.0	8.1	215	0.5
Bucaramanga	8.6	798.0	8.1	842	0.5
Ciénaga	8.4	234.0	7.9	252	0.5
Magangué	8.3	205.0	7.8	207	0.5

and here below ¿100

Table 4: Life satisfaction by municipality, where non-missing on life satisfaction ¿200 or 220??

municipality name	mean life satisfaction 2023	non missing life satisfaction 2023	mean life satisfaction 2021	non missing life satisfaction 2021	2023-2021 diff
Acacías	8.3	229.0	8.2	233.0	0.1
Aguachica	8.6	202.0	8.1	215.0	0.5
Aguadas	8.4	153.0	7.9	161.0	0.5
Aguazul	8.2	143.0	8.3	171.0	-0.1
Aguazul	8.0	102.0	8.3	171.0	-0.3
Aipe	8.1	112.0	8.0	100.0	0.1
Anserma	8.2	129.0	8.4	174.0	-0.2
Apartadó	8.3	119.0	8.2	111.0	0.1
Apartadó (Chigorodó)	7.8	132.0	NaN	NaN	NaN
Aracataca	8.3	190.0	7.7	122.0	0.6
Arauca	7.9	199.0	8.3	199.0	-0.4
Armenia	8.1	668.0	8.5	681.0	-0.4
Ataco	NaN	NaN	8.3	101.0	NaN
Barichara	8.0	131.0	7.9	128.0	0.1
Barrancabermeja	8.6	270.0	8.0	199.0	0.6
Barranquilla	8.3	1363.0	8.1	1116.0	0.2
Bello	NaN	NaN	8.5	103.0	NaN
Bello (Copacabana)	NaN	NaN	8.5	103.0	NaN
Bogotá, D.C.	8.2	3446.0	8.1	3836.0	0.1
Bosconia	8.3	106.0	7.7	124.0	0.6
Bucaramanga	8.6	798.0	8.1	842.0	0.5
Buenaventura	8.2	220.0	8.1	166.0	0.1
Buenaventura	8.0	192.0	8.1	166.0	-0.1
Buenaventura	8.2	220.0	8.3	279.0	-0.1
Buenaventura	8.0	192.0	8.3	279.0	-0.3
Buga	8.2	193.0	8.0	215.0	0.2
Cajibío	7.5	152.0	7.7	167.0	-0.2
Cajicá	NaN	NaN	7.8	106.0	NaN
Calamar	8.4	105.0	NaN	NaN	NaN
Calarcá	8.4	244.0	8.3	257.0	0.1
Cali	8.0	985.0	8.1	1108.0	-0.1
Candelaria	8.0	113.0	7.9	154.0	0.1
Cartagena	8.2	591.0	8.4	572.0	-0.2
Cartagena del Chairá	8.1	120.0	7.8	136.0	0.3
Cartago	7.6	168.0	7.5	187.0	0.1
Cereté	8.2	131.0	8.1	109.0	0.1
Chaparral	NaN	NaN	7.9	108.0	NaN
Chinchiná	8.5	141.0	8.3	198.0	0.2
Chiquinquirá	8.4	128.0	7.9	152.0	0.5
Circasia	8.0	102.0	NaN	NaN	NaN
Ciénaga	8.4	234.0	7.9	252.0	0.5
Codazzi	8.0	103.0	NaN	NaN	NaN
Corozal	8.1	174.0	7.7	167.0	0.4
Coyaima	NaN	NaN	8.1	110.0	NaN
Curillo	7.9	103.0	7.8	123.0	0.1
Cúcuta	8.3	573.0	8.3	553.0	0.0
Cúcuta (El Zulia)	NaN	NaN	8.1	106.0	NaN
Dosquebradas	8.0	103.0	8.5	141.0	-0.5
Duitama	8.0	134.0	8.2	138.0	-0.2
El Carmen de Bolívar	7.8	200.0	8.0	167.0	-0.2
El Doncello	8.3	158.0	8.0	160.0	0.3
El Paujil	8.0	100.0	7.9	106.0	0.1
Envigado	7.7	146.0	8.1	180.0	-0.4
Espinal	8.3	170.0	7.8	220.0	0.5
Facatativá	8.1	200.0	8.3	222.0	-0.2
Flandes	7.9	135.0	8.0	195.0	-0.1
Florencia	8.2	575.0	8.0	630.0	0.2
Florida	8.4	119.0	NaN	NaN	NaN
Fundación	8.3	200.0	7.8	155.0	0.5
Fusagasugá	8.2	156.0	7.3	117.0	0.9
Garagoa	NaN	NaN	7.9	109.0	NaN
Garzón	8.1	181.0	8.2	161.0	-0.1
Girardot	8.1	133.0	7.3	151.0	0.8
Granada	8.2	186.0	8.0	185.0	0.2
Guachené	7.3	138.0	7.7	161.0	-0.4
Guamo	NaN	NaN	8.1	106.0	NaN
Génova	NaN	NaN	8.3	101.0	NaN
Honda	NaN	NaN	8.0	102.0	NaN
Ibagué	8.4	478.0	8.2	549.0	0.2
Inírida	7.8	102.0	7.4	120.0	0.4
Inírida	7.8	102.0	7.5	149.0	0.3
Ipiales	7.9	137.0	7.1	134.0	0.8
Itagüí	NaN	NaN	8.4	128.0	NaN
Jamundí	NaN	NaN	8.1	103.0	NaN
La Calera	8.2	105.0	NaN	NaN	NaN
La Dorada	8.3	189.0	8.1	197.0	0.2
La Montañita	8.2	106.0	NaN	NaN	NaN
La Plata	7.7	133.0	7.9	128.0	-0.2
La Tebaida	NaN	NaN	8.7	113.0	NaN
Lorica	8.1	201.0	8.2	150.0	-0.1
Líbano	8.4	116.0	8.2	113.0	0.2
Magangué	8.3	205.0	7.8	207.0	0.5
Maicao	8.1	167.0	7.8	162.0	0.3
Malambo	8.3	234.0	7.8	136.0	0.5
Manizales	8.4	462.0	8.4	567.0	0.0
Manzanares	NaN	NaN	7.9	103.0	NaN
Mariquita	8.2	173.0	7.9	201.0	0.3
Marmato	NaN	NaN	8.5	121.0	NaN
Marulanda	NaN	NaN	8.5	100.0	NaN
Medellín	8.4	905.0	8.4	959.0	0.0
Mitú	8.2	173.0	7.7	181.0	0.5
Mocoa	8.2	112.0	7.4	128.0	0.8
Mompós	8.2	124.0	8.2	135.0	0.0
Montenegro	7.9	146.0	8.4	176.0	-0.5
Montería	8.3	465.0	8.0	372.0	0.3
Morelia	NaN	NaN	7.7	111.0	NaN
Natagaima	8.1	103.0	7.8	102.0	0.3
Neira	8.2	109.0	8.5	114.0	-0.3
Neiva	8.3	417.0	8.0	436.0	0.3
Ocaña	8.4	182.0	8.2	200.0	0.2
Paipa	NaN	NaN	7.9	104.0	NaN
Palmira	7.9	319.0	8.1	332.0	-0.2
Pamplona	8.5	130.0	8.3	116.0	0.2
Pasto	7.9	244.0	7.7	309.0	0.2
Paz de Ariporo	NaN	NaN	7.9	113.0	NaN
Pereira	8.4	524.0	8.3	677.0	0.1

7.3 Urban-Rural and Farmers

Colombia is quite urbanized at around 70-80%. https://population.un.org/wup/countryprofiles?country=Colombia&mainCategory Countries%20and%20Aggregates&subCategory=Urbanization%20-%20International%20definition

Still, a substantial part at 20-30% is rural.

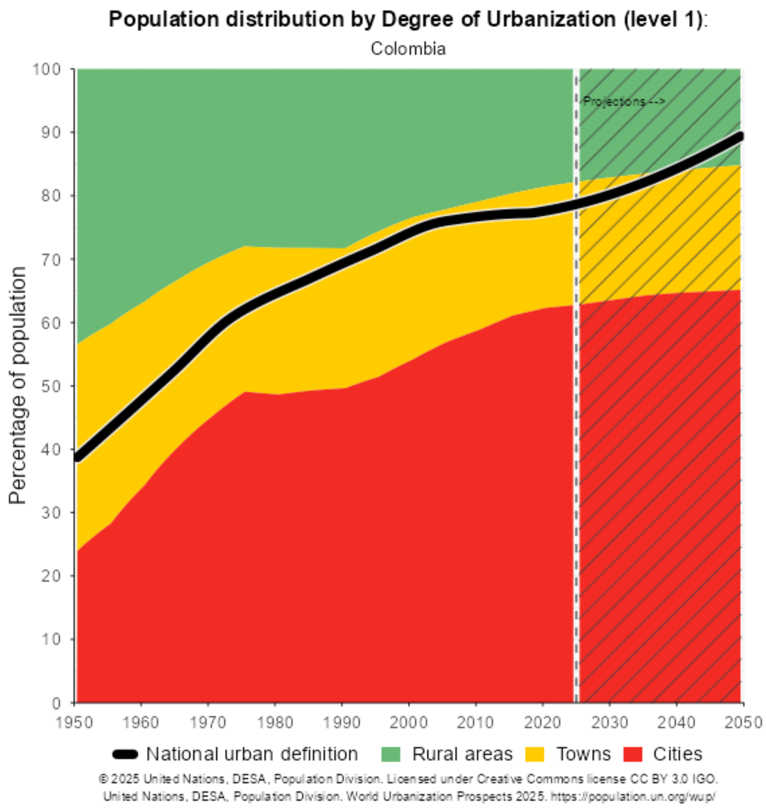


Figure 7

For rural in 2021 pattern is similar to 2023 in body of the paper but less strong for lower deciles where rural are happier than urban, but stronger for 2 top deciles where the opposite is true. For farmers in 2021 there is no happiness premium in lower deciles, but for non-farmers there is happiness premium in 2 top deciles.

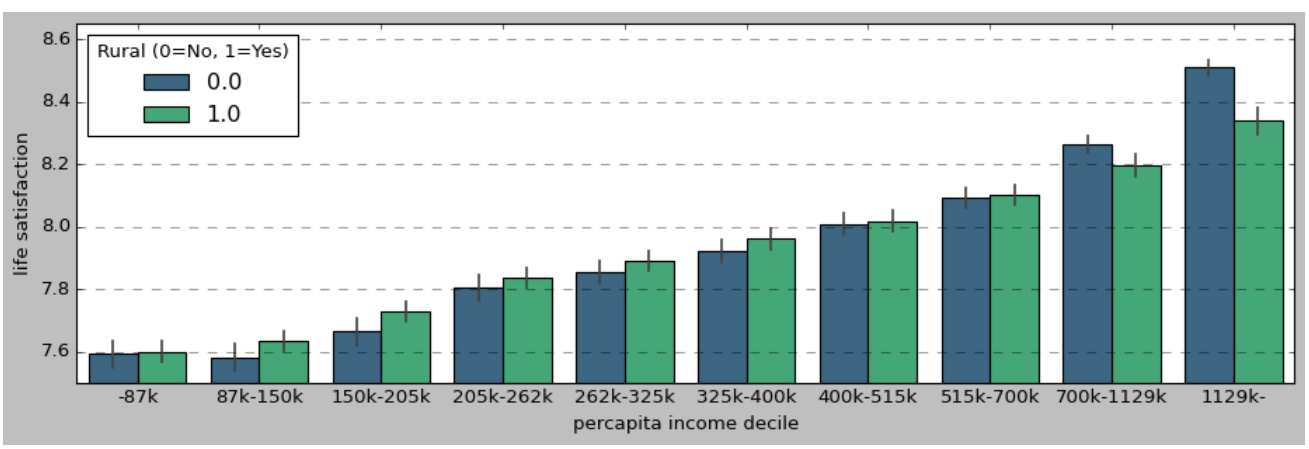


Figure 8

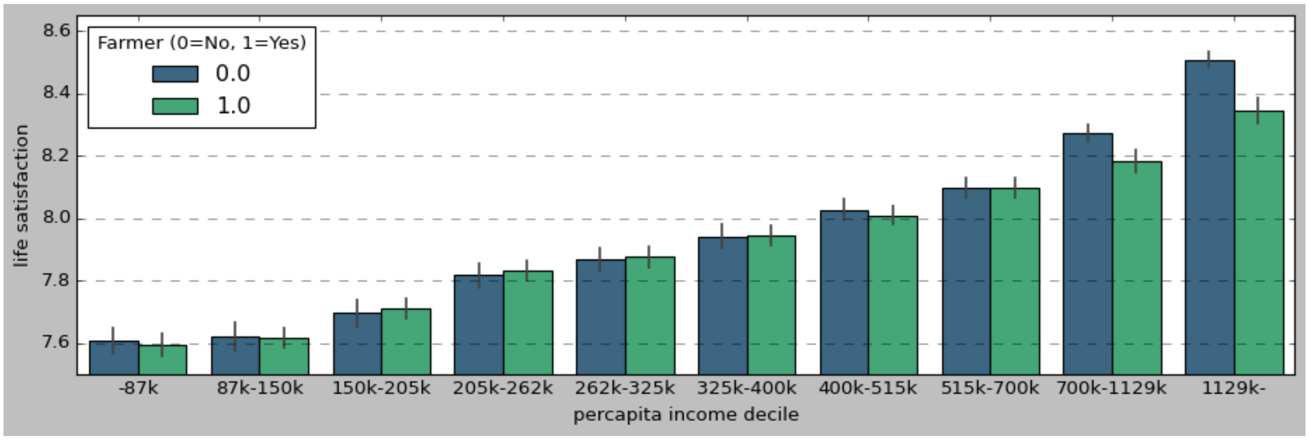


Figure 9

7.4 Geography of Colombia

The paper is about geography of happiness, but it would be incomplete without some mention of geography. First map of departments in figure **??**.

Figure 10: Colombia departments. Source: `https://upload.wikimedia.org/wikipedia/commons/thumb/6/67/Departments_of_colombia.svg/960px-Departments_of_colombia.svg.png`.

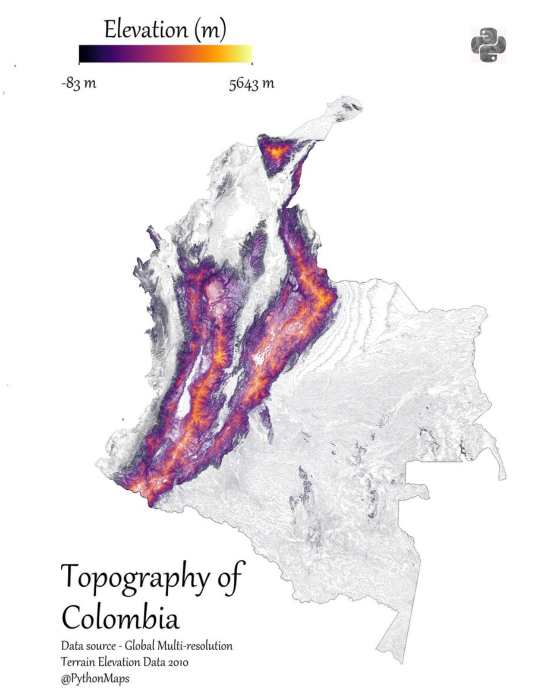
Topography is well mapped and best viewed at `https://en.wikipedia.org/wiki/File:Mapa_de_Colombia_(relieve).svg`, where you can zoom in to see the detail.

Population is nicely visualized in a 3d map at https://geoportal.dane.gov.co/geovisores/sociedad/grilla-dane-vihope/.

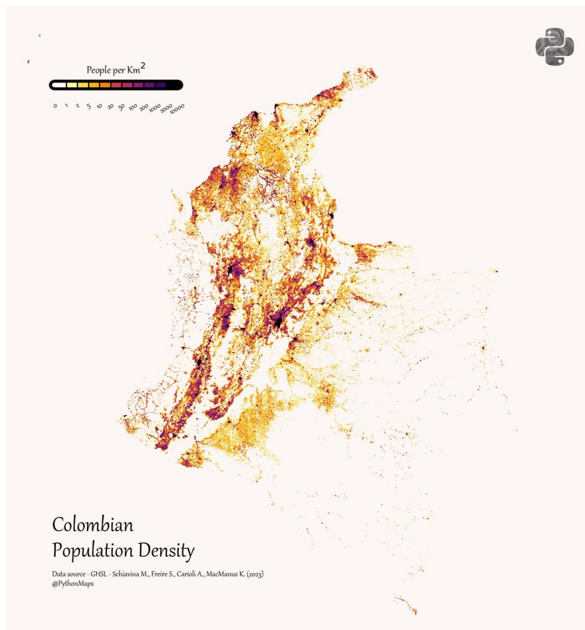
Topography v population density is visualized in figure 11. An interesting point about topography and population density is that much of the population is concentrated around montaneous areas (with a notable exception of largely flat Caraibean region). Plus that road infrastructure and connectivity is poor. Then this could be one explanation for large distinctivness/uniqness of regions–they were not homogenized as in many other countries as they are relatively weakly connected due to mountains and poor road infrastructure.

Figure 11: Topography and population density. Both maps come from visually appealing Python maps, and hi-res versions are at the below links. A 3d topography map is also at https://www.reddit.com/r/MapPorn/comments/u9gr9j/the_topography_of_colombia. And another topography map at https://upload.wikimedia.org/wikipedia/commons/3/3b/Colombia_Topography.png.

(a) Topography https://x.com/PythonMaps/status/1663937378782937088/photo/1



(b) Population density https://x.com/PythonMaps/status/1699089505133772803/photo/1



Land use is shown in **??**–note that most of the country is still natural. A clean historical 1970 map is at https://commons.wikimedia.org/wiki/File:Colombia_land_1970.jpg.

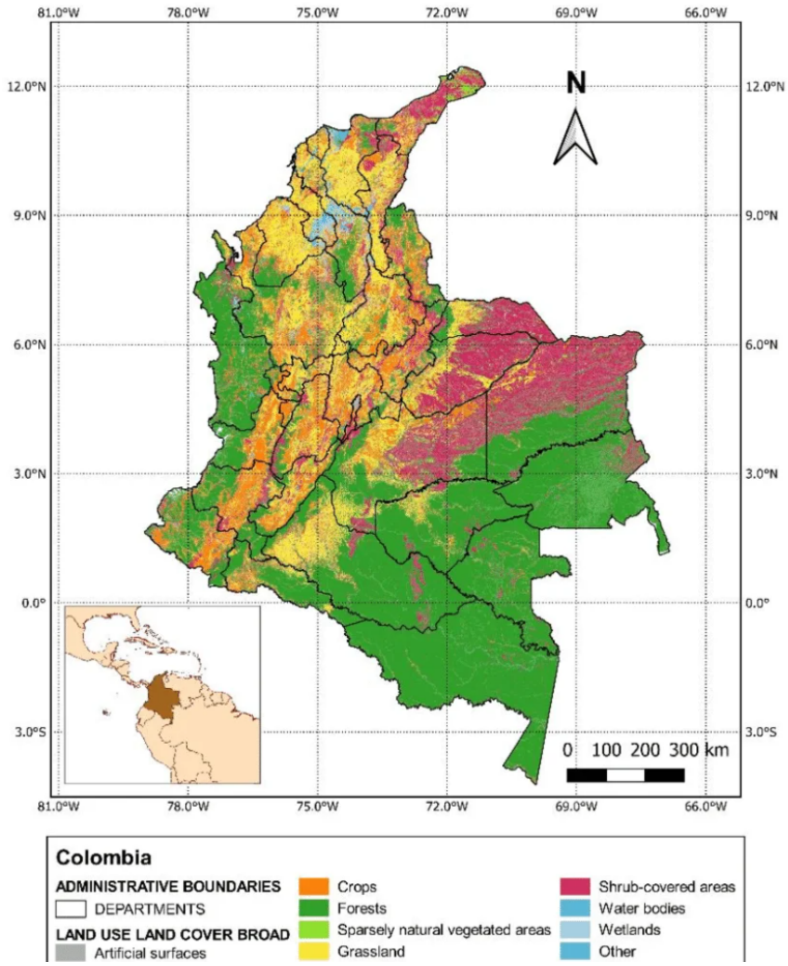


Figure 12: Land use. Source: `https://www.researchgate.net/figure/Land-use-and-land-cover-map-of-Colombia-2012_fig3_346648643`.

Likewise there is a lot of biomass in natural areas (and non-desert and not frozen) in figure **??**.

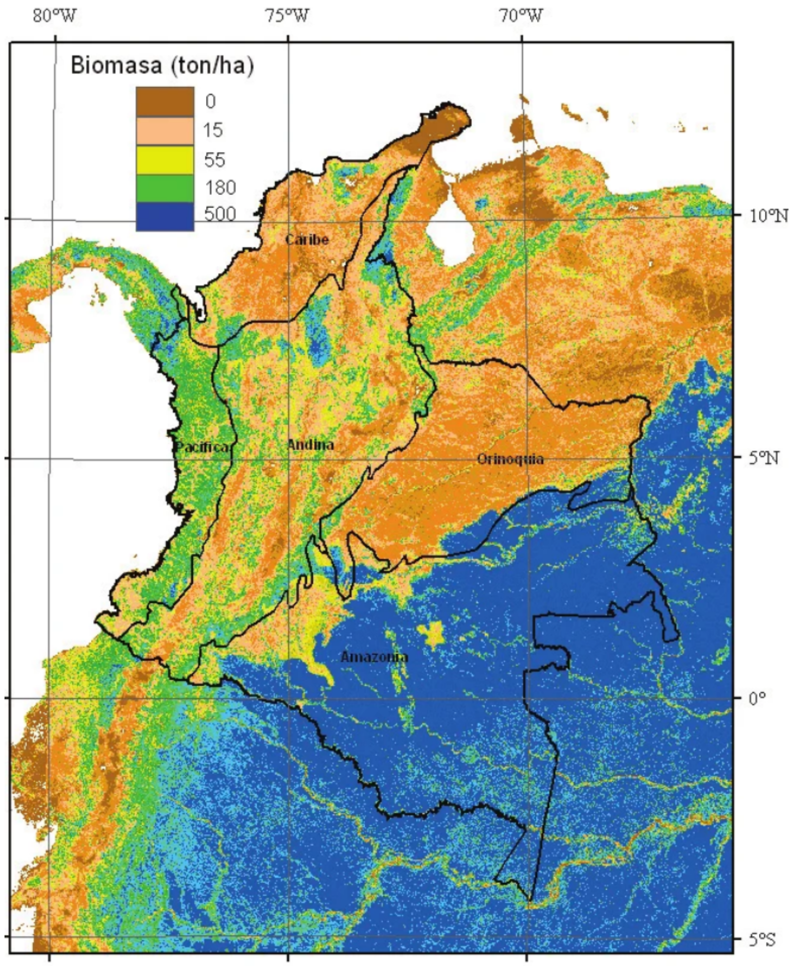


Figure 13: Biomass. Source: Anaya, J. A., Chuvieco, E., & Palacios, A. (2008). Estimación de biomasa aérea en Colombia a partir de imágenes MODIS. Revista de Teledetección, 30, 5-22. `https://www.researchgate.net/figure/Figura-7-Distribucion-espacial-de-valores-promedio-de-biomasa-en-Colombia_fig4_255626661`.

Last but not least ecological footprint. `https://data.footprintnetwork.org/` is a useful global overview, Colombian footprint per person is quite low at 2gha, 4x lower than that of the US at 8 gha, and it is stable since 1960s, but biocapcity per person has declined from 11.4 gha in 1960 to 3.3 in 2024. `https://happyplanetindex.org/countries/COL/` is another tool for cross-country comparisons.

`https://www.sciencedirect.com/science/article/pii/S1470160X20305677` provides useful hi-resolution visualizations of footprint across Colombia 1970-2015.

OECD provides more recent update across Colombia: `https://www.oecd.org/en/publications/oecd-environmental-performance 968398f7-en/full-report/biodiversity-conservation-and-sustainable-use_2099ac2c.html`

References

Glaeser, E. (2011): *Triumph of the City: How Our Greatest Invention Makes Us Richer, Smarter, Greener, Healthier, and Happier*, Penguin Press, New York NY.

Graham, C. (2009): *Happiness around the world: The paradox of happy peasants and miserable millionaires*, Oxford University Press, New York NY.

Okulicz-Kozaryn, A. and R. R. Valente (2021): "Urban unhappiness is common," *Cities*, 118, 103368.

O'Sullivan, A. (2009): *Urban economics*, McGraw-Hill.

Peiró-Palomino, J., A. J. Picazo-Tadeo, and E. Tortosa-Ausina (2020): "The Geography of Well-being in Colombia," Tech. rep.

PNUD (2023): *Informe sobre Desarrollo Humano para Colombia, cuaderno 2. Percepciones y bienestar subjetivo en Colombia: mas alla de indicadores tradicionales.*, https://www.undp.org/es/colombia/publicaciones/informe-desarrollo-humano-colombia-cuaderno-2.

Rojas, M. (2015): *Handbook of Happiness Research in Latin America*, Springer.

——— (2019): *Well-being in Latin America: Drivers and policies*, Springer Nature.